

A Short Empirical Investigation of the Traditional Phillips Curve Relationship in the UK

Introduction:

The Phillips curve is a well-established theory in Macroeconomics which suggests a negative relationship between Inflation and the level of unemployment. It holds a central role in macroeconomic thought and has guided policy for much of its existence. The purpose of this report is to investigate whether this relationship holds up based on the UK inflation and unemployment rate over time.

A Summary of the theories examined

The Phillips curve is a theorised negative relationship between Inflation and the rate of unemployment. Developed during the Keynesian era of economics where policy was predominantly focussed on demand management, it suggested that governments would face a trade-off between controlling inflation and reducing unemployment. For brevity, we will also establish that an alternative way the Phillips curve is formulated is instead as the relationship between the change in inflation and unemployment, rather than simply the level of inflation itself, sometimes referred to as the 'expectations-augmented Phillips curve'. This differs as it is based around the idea of the Non-accelerating inflationary rate of unemployment (NAIRU), as well as the idea that economic agents form expectations about future inflation that affect their behaviour. This model therefore postulates that unemployment is correlated negatively with inflationary pressures of the economy.

Methodology

To analyse the Phillips curve relationship, data from the ONS on the historical monthly UK unemployment rate and inflation rate as measured by the annual CPI measurement was gathered and imported into R. These

measures were picked for their simplicity in interpretation, with CPI being the preferred measurement for inflation due to its weighting of different consumer goods expenditure to provide an accurate basis for changes in the price level. The common period for the data was from 1989-2026 and so this was the period examined. From there, a simple bivariate regression and correlation analysis was applied to determine the relationship historically between unemployment and the inflation. Model 1 was used to analyse the relationship between the inflation rate as measured by the CPI and the rate of unemployment. For model 1, the equation used is

$$\pi_t = \beta_0 + \beta_1 u_t + \varepsilon$$

Model 1 was also broken down into 3 different periods. 1989-2007, 2008-2019, 2020-2026. The justification for this breakdown is to attempt to isolate large and abrupt changes in the economic conditions of the UK, with the financial crisis in 2007/8 and the COVID lockdown beginning in 2020.

Model 2 was then used to investigate if the relationship between unemployment and the change in inflation provided a more distinct correlation. The equation for model two was thus

$$\pi_t - \pi_{t-1} = \beta_0 + \beta_1 u_t + \varepsilon$$

with $\pi_t - \pi_{t-1}$ being defined as the difference in the annual inflation rate between consecutive months.

Finally, for our Hypothesis, we will establish the null hypothesis:

$$H_0: \beta = 0$$

And the alternative Hypothesis:

$$H_1: \beta \neq 0$$

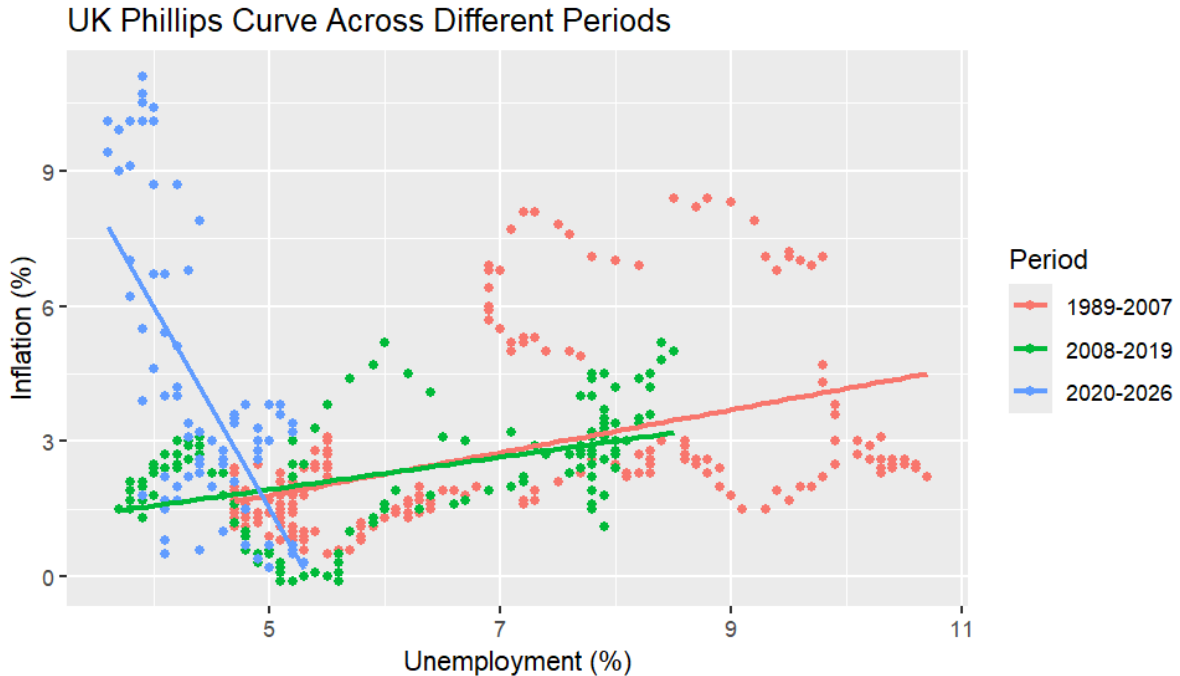
Results

	Value
Intercept (β_0)	1.94022
Std. Error β_0	0.35283
Unemployment coefficient (β_1)	0.14476

Std. Error β_1	0.05464
R-squared	0.01543
Adjusted R-squared	0.01323
P value	0.008346

Table 1. Regression results and coefficients (Model 1)

Model 1 initially suggests a relationship counterintuitive to what we expect, with instead a positive correlation between unemployment and inflation. While statistically significant at the 1% level, the R^2 value is only 0.01543, meaning unemployment only explains 1.54% of the variations in inflation. This combined with a weak regression coefficient, only around 0.145, means it holds little explanatory power. However, when we break up model 1 into the respective time periods we defined earlier, then the regression coefficients suggest the Phillips relationship may operate differently under different economic conditions.



Graph 1. Regression Results by Period (Model 1)

	1989–2007	2008–2019	2020–2026
Intercept (β_0)	-0.55901	0.13945	23.83132

Std. Error β_0	0.44629	0.37668	2.63062
Unemployment coefficient (β_1)	0.47406	0.35928	-4.46429
Std. Error β_1	0.06295	0.05946	0.59468
P value	<0.001	<0.001	<0.001
R-squared	0.20060	0.20453	0.42579
Adjusted R-squared	0.19706	0.19892	0.41824

Table 2. Regression Results by Period (Model 1)

For the periods 1989-2007 and 2008-2019, there were stronger positive relationships between inflation and unemployment (0.474 and 0.359 respectively) compared to the aggregate model 1 result of 0.145. By contrast the period 2020-2026 saw a much more traditional Phillips curve relationship, with a coefficient of -4.464. Given the context of the Phillips curve emerging in the context of Keynesianism and demand management, it is possible the UK's economic conditions in this time period matched more closely with the problems of the Keynesian era than other time period and hence showed a stronger Philips curve relationship, however other factors such as the supply disruptions and trade shocks of the time period would need to be studied alongside it to fully establish the reasons for why the relation holds stronger in this period compared to the others. Furthermore, the regression results here alone cannot establish that the relationship changed because of any particular economic conditions.

	Value
Intercept (β_0)	0.109293
Std. Error β_0	0.061481
Unemployment coefficient (β_1)	-0.018482
Std. Error β_1	0.009525
P value	0.052972

R-squared	0.008352
Adjusted R-squared	0.006134

Table 3. Regression Results and coefficients (Model 2)

Model 2, our investigation of the expectations augmented version of the Phillips curve, shows a negative, yet still exceptionally weak, relationship between unemployment and the rate of change of inflation. Notably however, the P value is now 0.05297, meaning we would not reject the null hypothesis at the 5% significance level, whereas in model 1 in both the aggregate and individual periods, we would reject the null hypothesis at the 1% level.

Conclusions

The results show that while a Phillips curve relationship may not be necessarily obvious in aggregate data across the time-period, the Phillips curve relationship may be more apparent under certain economic conditions and periods. Additionally, the evidence suggests an, albeit weak, Phillips curve relationship when examining it under the formulation of the expectations-augmented framework. With such generally small coefficients however, the findings should be interpreted as weak evidence rather than confirmation of a stable Phillips curve relationship.

It is important to realise the limited data used in this investigation, as well as the role of other external forces in influencing inflation and unemployment rates such as foreign markets, time lags, irrational behaviour, government policy interaction, etc. A more thorough investigation with data to account for other explanatory variables may be able to isolate such effects and reach a more precise conclusion on the exact Phillips curve relationship in the UK and under what conditions the relationship is more or less pronounced. Such investigations are beyond the capability of a simple bivariate regression with standard OLS assumptions